Standard Article Template

Testing the Native LaTeX Renderer

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1 Introduction

This document tests the native LaTeX{} renderer across a wide range of constructs.
We include **colored text**, mathematics, tables with `booktabs`,
code listings, custom theorem environments, and `tcolorbox` environments.

The renderer should handle all of the following:

\Roman1. Mathematical typesetting (inline and display)
\Roman2. Custom macro expansion
\Roman3. Color support via `xcolor`
\Roman4. Table environments with `booktabs`
\Roman5. Code listings
\Roman6. Box environments via `tcolorbox`

2 Mathematics

2.1 Inline Mathematics

Einstein's famous equation *E = mc^2* appears inline. The quadratic formula gives *x = frac{-b ± sqrt{b^2 - 4ac}}{2a}* for roots of *ax^2 + bx + c = 0*.

2.2 Display Mathematics

The Fourier transform of a function *f in L^1(mathbb{R})* is defined as:

$$\hat{f}(\xi)=\int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x\xi}\, dx$$

$$\nabla\cdot\mathbf{E}\&=\frac{\rho}{\varepsilon_0}\backslash\backslash\ \nabla\cdot\mathbf{B}\&=0\backslash\backslash\ \nabla\times\mathbf{E}\&=-\frac{\partial\mathbf{B}}{\partial t}\backslash\backslash\ \nabla\times\mathbf{B}\&=\mu_0\mathbf{J}+\mu_0\varepsilon_0\frac{\partial\mathbf{E}}{\partial t}$$

Maxwell's equations (??)–(??) govern classical electromagnetism.

[Continuity]
A function f: mathbb{R}to mathbb{R} is continuous at x_0 if for every varepsilon > 0 there exists δ > 0 such that left | x - x_0right | < δ implies left | f(x) - f(x_0)right | < varepsilon.

[Intermediate Value Theorem]

If f: [a,b] to mathbb{R} is continuous and y is between f(a) and f(b), then there *exists c in (a,b) with f(c) = y*

1	This is a longer description that should wrap within the X column	Active	42
2	Another item with a description	Inactiv	17
3	Third entry	Active	99

Table: A tabularx table with automatic column widths

4 Color and Boxes

Text can be red, blue, or dark green. You can also use highlighted backgrounds.

Important Note

Warning

5 Code Listings

```
, label=lst:render]
func typeset(_ blocks: [TexBlock]) -> TypesetDocument {
    var pages: [TypesetPage] = [TypesetPage(pageNumber: 1)]
    var columnIndex: Int = 0
    var cursorY: CGFloat = geometry.marginTop

    for block in blocks {
        switch block {
        case .paragraph(let inlines):
            let attrStr = makeParaAttrStr(inlines, fonts: fonts)
            appendTextBlock(attrStr, pages: &pages, ...)
        case .mathDisplay(let src):
            let node = MathParser.parse(src)
            // ... render math
        default:
            break
        }
    }
    return TypesetDocument(pages: pages, ...)
}
```

Inline code: `TeXTypesetter.typeset(\_:)` is the main entry point.

6 Lists with Enumitem

$\checkmark$ First item with check mark

$\checkmark$ Second item

$\checkmark$ Third item

(c) This list starts at (c)

(d) Continues with (d)

(e) And so on

7 Cross-References

We defined the Fourier transform in Eq. (??).
Maxwell's equations are in Eq. (??)–(??).
The Intermediate Value Theorem is Theorem ??.
Results are in Table ??.

8 Figures

Figure ?? illustrates the rendering pipeline. Images are loaded via `CGImageSource` and scaled to fit the column width.

Figure 1. A sample figure demonstrating \textbackslash includegraphics support. The image is automatically scaled to 70% of text width.

9 Bibliography

Prior work by LeCun et al. [1] established deep learning as a general-purpose feature extractor. The attention mechanism [2] underpins modern large language models. Classical typesetting follows Knuth [3]. Online collaborative tools like Overleaf [4] have changed how papers are written.

10 Conclusion

This document exercises most of the packages and constructs needed for real-world academic papers. Any construct not rendered correctly represents a gap to close in the parity effort.